



Portfolio

MUHAMMAD NASHIRUDDIN AL BANI

Data Enthusiast

mnashiruddin.albani23@gmail.com



[linkedin.com/in/mhmd-al-bani](https://www.linkedin.com/in/mhmd-al-bani)

ABOUT ME



Muhammad Nashiruddin Al Bani

My name is Muhammad Nashiruddin Al Bani. I am a fresh graduated from Yogyakarta State University with a Bachelor's degree in Statistics. During my time as a student, I was actively involved in various organizations and projects, including the Student Executive Board and community-based initiatives in Yogyakarta. I also took part in several competitions. I have a strong interest in research, statistical consulting, data analysis, and mathematics.

mnashiruddin.albani23@gmail.com

Education



- 2021 - 2025

Yogyakarta State University

Statistics Study Program

GPA 3.70/4.00



Forage

- August 2024

**PwC Switzerland Power BI
Job Simulation on Forage**

mnashiruddin.albani23@gmail.com

Skills & Tools



Technical Skills

- Web-Based Information System Development
- Database Design and Management
- Data Analysis and Statistical Modeling
- Data Mining and Exploratory Data Analysis
- Machine Learning and Clustering Analysis
- Artificial Intelligence and Computer Vision
- Digital Image Processing and Optical Character Recognition (OCR)
- Data Visualization and Interactive Dashboard Development
- Linux Server Deployment and Administration
- Data Automation using Google Apps Script
- Administrative Digitalization and Technical Documentation



Soft Skills

- Team Leadership
- Problem Solving
- Professional Communication
- Team Collaboration
- Project Planning
- Analytical Thinking
- Time Management
- Technical Documentation



Tools



Professional Experience 1



**Pusat Pelatihan BPSDM Imigrasi dan Pemasyarakatan,
Kementerian Imigrasi dan Pemasyarakatan Republik Indonesia**

November 2025 - May 2026

***Training Center of the Human Resources Development Agency for
Immigration and Corrections,
Ministry of Immigration and Corrections of the Republic of Indonesia***

- **Internship**



Technical Training Data Management Intern at the Training Center of the Human Resources Development Agency for Immigration and Corrections

- Managed, validated, and processed nationwide technical training data for Immigration and Corrections personnel.
- Developed and maintained structured training databases to support strategic HR development planning and evaluation.
- Conducted statistical analysis to measure training effectiveness based on key performance indicators and participant evaluation results.
- Prepared analytical and statistical reports to support managerial decision-making.
- Coordinated with the Competency Development and Assessment Center to ensure data accuracy, completeness, and security compliance.

Professional Experience 1



**Pusat Pelatihan BPSDM Imigrasi dan Pemasyarakatan,
Kementerian Imigrasi dan Pemasyarakatan Republik Indonesia**

November 2025 - May 2026

***Training Center of the Human Resources Development Agency for
Immigration and Corrections,
Ministry of Immigration and Corrections of the Republic of Indonesia***

- **Tools**



Google Apps Script



- **Skills**

1. Data Management
2. Data Processing
3. Database Management
4. Database Development using Google Apps Script
5. Advanced Microsoft Excel (Formulas, Pivot Tables, Data Validation, XLOOKUP/VLOOKUP)
6. Data Analysis
7. Data Validation and Data Quality Control
8. Data Processing Automation
9. Training Database Administration
10. Digital Data Documentation

Professional Experience 1



**Ketua Tim Pengembangan dan Implementasi Sistem Rumah Tangga
Badan Pengembangan Sumber Daya Manusia Imigrasi dan Pemasyarakatan
Kementerian Imigrasi dan Pemasyarakatan Republik Indonesia**

November 2025 - May 2026

***Team Leader for Household System Development and Implementation of the Human Resources Development Agency for Immigration and Corrections
Ministry of Immigration and Corrections of the Republic of Indonesia***

• Internship **Team Leader for Household System Development and Implementation of the Human Resources Development Agency for Immigration and Corrections**

Led a development team in designing and implementing a web-based information system to support administrative management within the Household Division of the Human Resources Development Agency for Immigration and Corrections. The system covered inventory management, item request processing, procurement records, and role-based access control for administrators and users. I was responsible for system development, Laravel application deployment on a Linux server, Nginx, PHP-FPM, and MariaDB configuration, administrative data processing and validation, structured database development, server troubleshooting, firewall configuration, and frontend asset optimization using Vite.

- Led the system design and development process collaboratively with the development team.
- Implemented a web-based inventory and item request management system.
- Deployed the application on the institution's internal production server.
- Developed a Role-Based Access Control (RBAC) mechanism.
- Resolved technical issues related to server configuration, permissions, databases, and frontend asset builds.
- Supported the digitalization of administrative processes within the Household Division of BPSDM IMIPAS.

Professional Experience 1



Ketua Tim Pengembangan dan Implementasi Sistem Rumah Tangga
Badan Pengembangan Sumber Daya Manusia Imigrasi dan Pemasyarakatan
Kementerian Imigrasi dan Pemasyarakatan Republik Indonesia

November 2025 - May 2026

*Team Leader for Household System Development and Implementation of the Human Resources Development Agency for Immigration and Corrections
Ministry of Immigration and Corrections of the Republic of Indonesia*

• Tools



• Skills

1. Web-Based Information System Design
2. System Development Team Leadership
3. User Requirement Analysis
4. Database Management
5. Data Processing and Validation
6. Linux Server Deployment and Configuration
7. Role-Based Access Control
8. System and Server Administration
9. Web Application Troubleshooting
10. Administrative Digitalization and Data Documentation

• Project Achievement

Successfully designed and implemented the BPSDM IMIPAS Household Information System to support item request processing, inventory management, procurement, and administrator-user access control. The system has been deployed on an internal server and is ready for further integration with the official domain.

Professional Experience 2



Dinas Kesehatan Kota Yogyakarta **Yogyakarta City Health Service**

July - August 2023

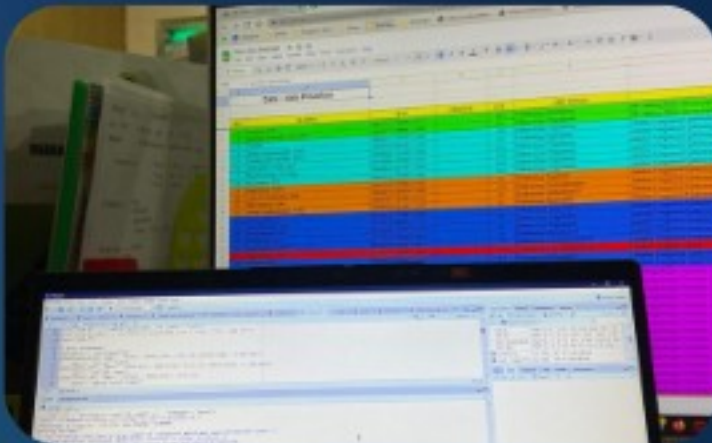
● Internship



Section for Prevention and Control of Non-Communicable Diseases and Mental Health

My duties include collecting and analyzing data related to non-communicable diseases and mental health, solving problems through data collection and analysis in various programs (including integrated coaching posts), and providing insights, recommendations, and suggestions on sustainability policies and strategies in health services and collaborating institutions in Yogyakarta.

● Internship



Pharmaceutical Section of Medical Devices and Food and Beverages

My duties include collecting and analyzing data related to the pharmaceutical, medical devices, and food and beverage sectors, resolving issues related to sampling of incoming and outgoing medical devices, conducting sampling at the Integrated Pharmaceutical and Medical Devices Service Unit Warehouse, and providing input, recommendations, and suggestions related to sustainability policies and strategies in the health services sector and collaborating institutions in the City of Yogyakarta.

● Internship



Health Human Resources Section

My job is to collect and analyze data related to health human resources, including pre-test and post-test data for employees and business premises applying for permits, evaluate, interpret, and present quantitative data, and provide input, recommendations, and suggestions on sustainability policies and strategies in health services and collaborating institutions in the City of Yogyakarta.

Professional Experience2



Dinas Kesehatan Kota Yogyakarta
Yogyakarta City Health Service

July - August 2023

- **Tools**



- **Skills**

1. Data Analysis and Statistical Analysis
2. Data Cleaning and Data Preprocessing
3. Exploratory Data Analysis (EDA)
4. Data Visualization and Dashboard Development
5. Statistical Modeling and Data Interpretation
6. Public Health Data Analysis
7. Data Mining and Pattern Identification
8. Data Reporting and Insight Presentation
9. Research and Data Documentation
10. Evidence-Based Data Interpretation for Decision-Making

Project 1

Training Center of the Human Resources Development Agency for Immigration and Corrections Ministry of Immigration and Corrections of the Republic of Indonesia

WEB-BASED TRAINING DATABASE INFORMATION SYSTEM USING GOOGLE APPS SCRIPT

Project Overview :

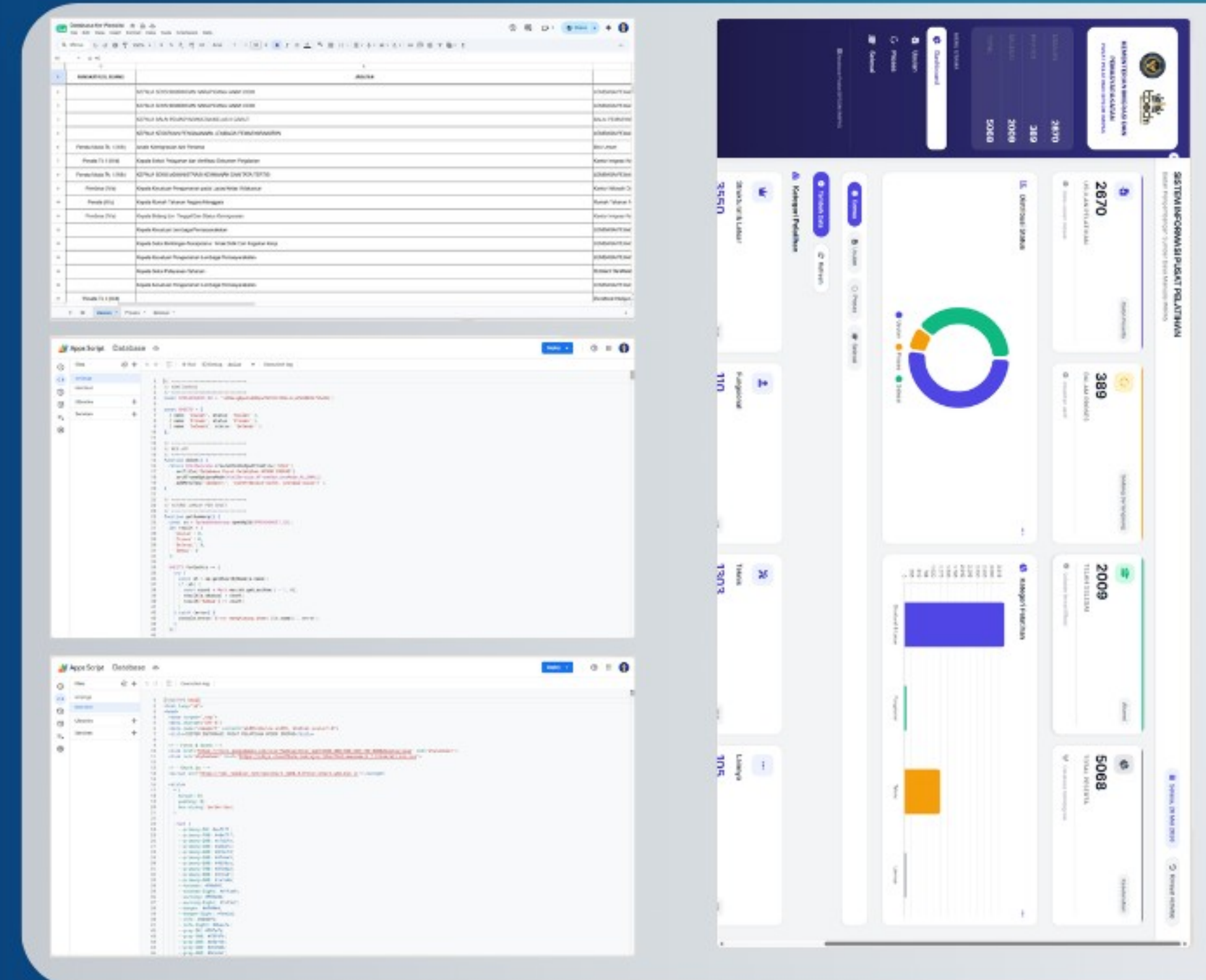
This project was developed to design and implement a web-based training database information system for the Training Center of BPSDM IMIPAS by integrating Google Spreadsheet and Google Apps Script to manage training participant data across proposed, ongoing, and completed training stages. The platform provides an interactive dashboard that presents participant summaries, training status distribution, training category classification, and detailed participant records. As the team leader, I coordinated the development process, designed the system workflow, structured the database, and ensured that the web application effectively supported data monitoring and administrative decision-making.

Technology & Tools :

- Programming Language: JavaScript
- Platform: Google Apps Script
- Database: Google Spreadsheet
- Frontend: HTML, CSS, JavaScript
- Data Visualization: Chart.js
- UI Framework: Custom Responsive Dashboard
- Deployment: Google Apps Script Web App

Model & Architecture

- Web-based dashboard integrated with Google Spreadsheet as the main database.
- Data classification based on training status: Proposed, In Process, and Completed.
- Training category clustering, including leadership training, functional training, technical training, and other training categories.
- Interactive data visualization using summary cards, charts, filters, and detailed tables.
- Search, filtering, and participant detail features to support administrative monitoring.
- Role as team leader in system planning, workflow design, data structuring, and implementation coordination.



https://github.com/Albani258/Portofolio_Muhammad-Nashiruddin-Al-Bani/tree/main/Project_01_Training_Database_Information_System_Google_Apps_Script

Project2

Team Leader for Household System Development and Implementation of the Human Resources Development Agency for Immigration and Corrections *Ministry of Immigration and Corrections of the Republic of Indonesia*

WEB-BASED HOUSEHOLD AND ITEM REQUEST MANAGEMENT INFORMATION SYSTEM

Project Overview :

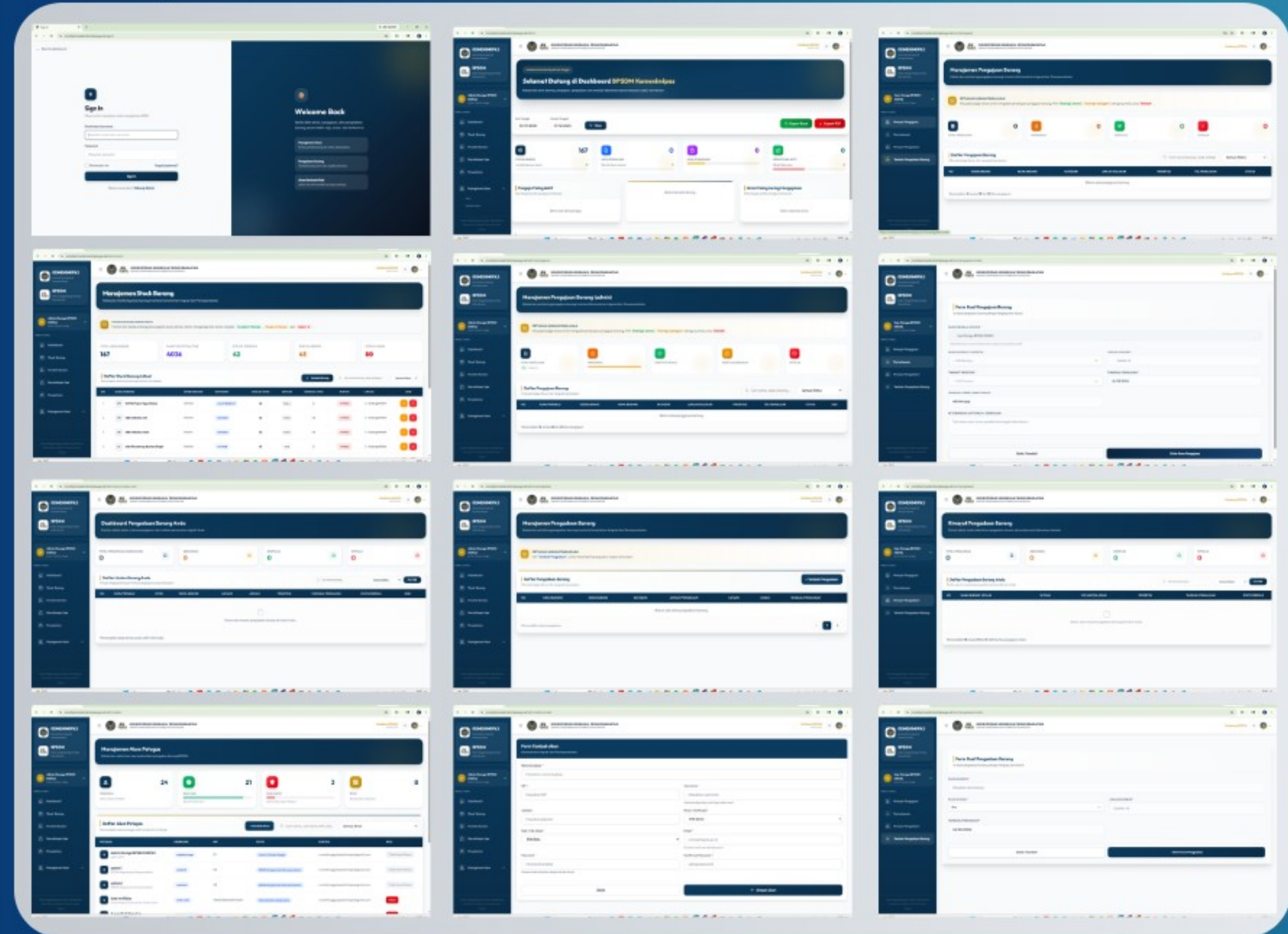
This project was developed to design and implement a web-based information system that supports administrative operations within the Household Division of BPSDM IMIPAS, particularly in inventory management, item request submissions, procurement processes, and role-based user management. The system was collaboratively developed with the development team, where I served as the team leader responsible for coordinating system development and implementation.

Technology & Tools :

- Programming Language: PHP
- Framework: Laravel
- Database Management System: MariaDB / MySQL
- Web Server: Nginx
- Runtime Service: PHP-FPM
- Frontend Build Tool: Vite
- Deployment Tools: PowerShell, SSH, Linux Server

Model & Architecture

- Web-based information system for inventory and item request management.
- Role-Based Access Control (RBAC) for administrator and user access management.
- Structured database design for inventory records, procurement management, user accounts, and item request processing.
- Linux server deployment architecture using Laravel, Nginx, PHP-FPM, and MariaDB.
- Internal server implementation with firewall configuration and readiness for official domain integration.



Dinas Kesehatan Kota Yogyakarta
Yogyakarta City Health Service

Project Overview :

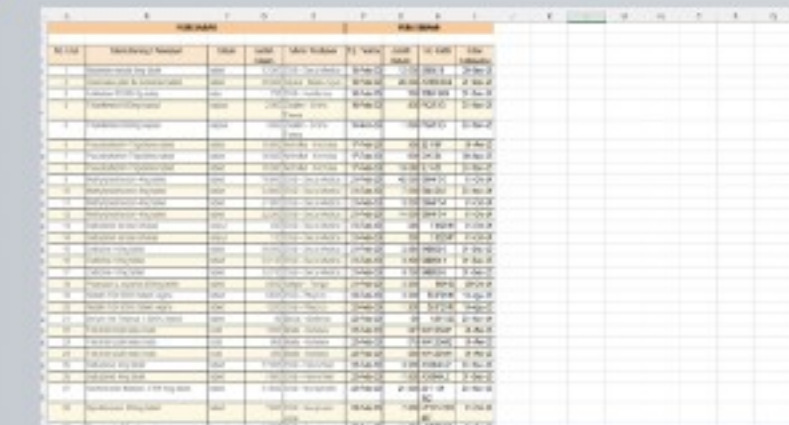
This project aims to analyze and cluster data using the K-Means method to identify patterns within the dataset. By leveraging data analytics technology, this project helps in grouping data based on similar characteristics, making it useful for more effective decision making.

Technology & Tools :

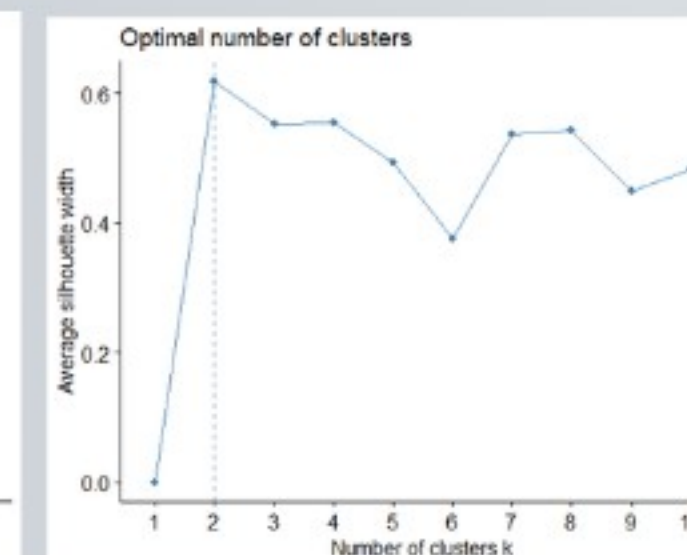
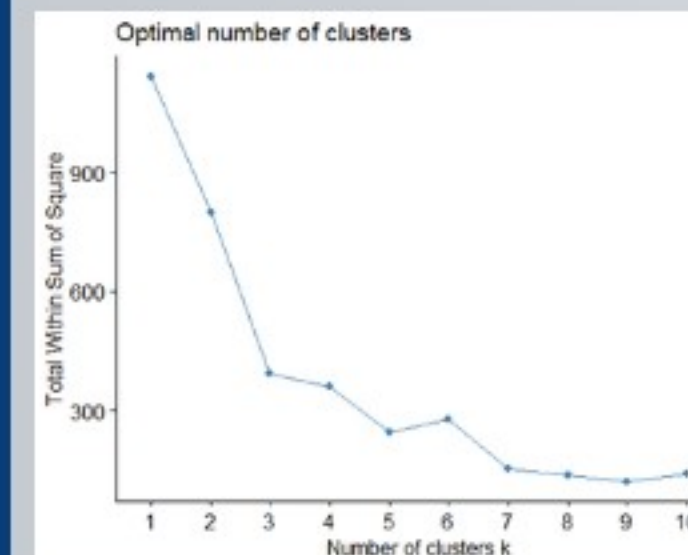
- Programming Language: R
- Data Processing: RStudio
- Visualization: ggplot2, base R plotting
- Machine Learning & Statistical Methods: K-Means Clustering

Model & Architecture

- Machine Learning Model: K-Means Clustering for data segmentation.
- Feature Engineering: Data preparation, descriptive analysis, determining the optimal number of clusters using the Elbow Method, initial cluster centroid analysis, centroid updating, object clustering, and model evaluation using validation metrics.

[illegible]

```
> # Menormalisasi data
> obafix <- scale(obat[,2:4])
>
> # Menghitung jarak Euclidean
> dataeuc <- get_dist(obafix)
>
> # Metode elbow untuk menentukan jumlah cluster optimal
> fviz_nbclust(obafix, kmeans, method = "wss")
>
> # Metode silhouette untuk menentukan jumlah cluster optimal
> fviz_nbclust(obafix, kmeans, method = "silhouette")
```

[illegible][illegible][illegible]

Project 4

Data Mining for Statistics

Customer Segmentation Analysis for Credit Card Holders Using Hierarchical Clustering

Project Overview :

This project aimed to group 8,651 active credit card holders (over 6 months) into meaningful customer segments using Hierarchical Clustering (Ward's Method). The dataset contained 18 behavioral variables, such as purchase frequency, one-off purchases, installment purchases, cash advances, credit limit, and payments.

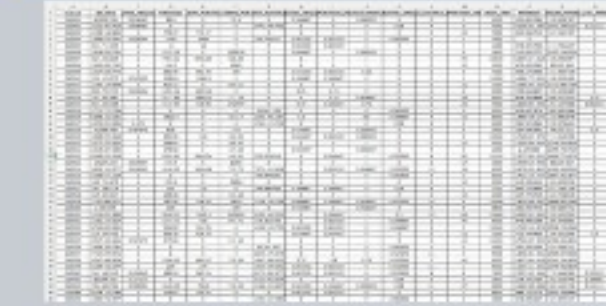
The objective was to identify spending patterns and segment customers to support targeted marketing strategies for financial institutions.

Technology & Tools :

- Programming Language: Python
- Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- Methods: Hierarchical Clustering (Ward's Method), PCA, Silhouette Score

Model & Architecture

- Preprocessing: Missing value imputation, feature scaling
- Best Cluster Number: 3 (evaluated with Silhouette Score)
- Cluster Insights:
 - Cluster 0: High balance & high spending customers (potential premium targets)
 - Cluster 1: Low balance & low spending (dormant users)
 - Cluster 2: Medium balance & medium spending (average customers)
- Visualization: Dendrogram, PCA-based scatter plots



```
import numpy as np #Memanggil data yang dibutuhkan array
import pandas as pd #Memanggil file csv dan display dalam bentuk tabel
import matplotlib.pyplot as plt #Digunakan untuk visualisasi data
from scipy.sparse import data

# Memanggil file dalam bentuk Diagram Batang
import seaborn as sns
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage

# Memanggil Silhouette Score
from sklearn.metrics import silhouette_score # Digunakan untuk mengukur kualitas cluster yang terbentuk

# Memanggil Standard Scaler
from sklearn.preprocessing import StandardScaler

# Mengurangi dimensi data sehingga dapat dilakukan visualisasi 2 dimensi
from sklearn.decomposition import PCA
from sklearn.metrics.pairwise import cosine_similarity
```

```
dataset = pd.read_csv('CC_GENERAL.csv')
print(dataset)

CREDIT_LIMIT  BALANCE  BALANCE_FREQUENCY  PURCHASES
0  C10001  49.300749  0.618182  95.40
1  C10002  3282.467816  0.989891  0.00
2  C10003  2496.148862  1.000000  773.17
3  C10004  1666.679542  0.424264  1499.00
4  C10005  817.714335  1.000000  14.00
...
```

```
# Menghapus kolom credit_id
dataset.drop('CREDIT_ID', axis=1, inplace=True)

# Mengkonversi nilai null (kosong) pada dataset
dataset.isnull().sum()
MINIMUM_PAYMENTS  303
CREDIT_LIMIT      1
BALANCE           0
CASH_ADVANCE_FREQUENCY  0
PNC_FULL_PAYMENT  0
PAYMENTS          0
dtype: object
```

```
dataset['CREDIT_LIMIT'] = dataset['CREDIT_LIMIT'].fillna(dataset['CREDIT_LIMIT'].mean())
dataset['MINIMUM_PAYMENTS'] = dataset['MINIMUM_PAYMENTS'].fillna(dataset['MINIMUM_PAYMENTS'].mean())

len(dataset.columns[dataset.isnull().any()])
dataset.isnull().sum().sort_values(ascending=False).head()
BALANCE  0
CASH_ADVANCE_FREQUENCY  0
PNC_FULL_PAYMENT  0
MINIMUM_PAYMENTS  0
PAYMENTS  0
dtype: int64
```

```
# Mengkonversi nilai
encoder = StandardScaler()
encoded_data = encoder.fit_transform(dataset)

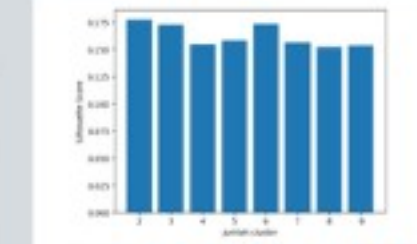
df = pd.DataFrame(encoded_data, columns=dataset.columns)
df.head()
```

```
# Standardisasi nilai
encoder = StandardScaler()
encoded_data = encoder.fit_transform(dataset)

df = pd.DataFrame(encoded_data, columns=dataset.columns)
df.head()
```



```
# 2. Silhouette Score
silhouette_score = 0
for k in range(1, 10):
    agglomerative_clustering = AgglomerativeClustering(n_clusters=k)
    silhouettes = []
    for i in range(0, len(encoded_data)):
        sil = agglomerative_clustering.fit_predict(encoded_data)
        sil_score = silhouette_score(encoded_data, sil)
        silhouettes.append(sil_score)
    plt.plot(range(1, 10), silhouettes)
    plt.xlabel('Number of Clusters')
    plt.ylabel('Silhouette Score')
```



```
# 3. Agglomerative Clustering dengan jarak cluster = 3
'''
Diketahui data clustering menggunakan metode Hierarchical Clustering dengan jarak cluster 3.
'''
agg = AgglomerativeClustering(n_clusters=3)
agg.fit(encoded_data)
label = agg.labels_
label_agg = pd.Series(label, index=encoded_data.index)
agg.get_labels()
```

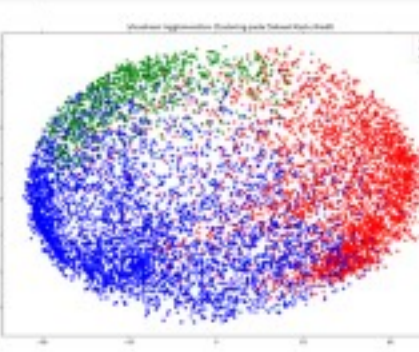
```
# Mengkonversi data ke format numpy array
dataset = dataset.to_numpy()
dataset = dataset.astype('float32')
```



```
# 5. Silhouette Score
silhouette_score = 0
for k in range(1, 10):
    agglomerative_clustering = AgglomerativeClustering(n_clusters=k)
    silhouettes = []
    for i in range(0, len(encoded_data)):
        sil = agglomerative_clustering.fit_predict(encoded_data)
        sil_score = silhouette_score(encoded_data, sil)
        silhouettes.append(sil_score)
    plt.plot(range(1, 10), silhouettes)
    plt.xlabel('Number of Clusters')
    plt.ylabel('Silhouette Score')
```

```
for name, group in groups:
    ax.scatter(group.x, group.y, marker='o', linestyle='', s=50,
              color = colors[name], label = label_pos[name], name = name)

ax.legend()
ax.set_title('Visualisasi Agglomerative Clustering pada Dataset Paris Dredge')
plt.show()
```



Project5

Digital Image Processing

Implementation of Vehicle License Plate Recognition using OpenCV and OCR in Automated Parking System

Project Overview :

Developed an automatic parking system with license plate detection & recognition. Combined OpenCV (image processing) and EasyOCR (text extraction). Tested on 15 real images → 83.85% accuracy.

Technology & Tools :

- Programming: Python
- Libraries: OpenCV, EasyOCR, NumPy, Matplotlib
- Methods: Image Processing, Segmentation, Optical Character Recognition

Model & Architecture

- Computer Vision Model: OpenCV for image preprocessing, plate detection, and segmentation.
- Text Recognition Model: EasyOCR for extracting alphanumeric characters from detected license plates.
- Feature Engineering: Image preprocessing (grayscale conversion, resizing, noise reduction, edge detection), plate localization using contour detection, character segmentation, and text conversion into ASCII.
- Evaluation: Accuracy measurement by comparing detected plate numbers against ground truth labels, with an overall accuracy of 83.85%.

```
1 import cv2
2
3 harcascade = "models/harcascade_russian_plate_number.xml"
4
5 cap = cv2.VideoCapture(0)
6
7 cap.set(3, 640) # width
8 cap.set(4, 480) # height
9
10 min_area = 500
11 count = 0
12
13 while True:
14     success, img = cap.read()
15
16     plate_cascade = cv2.CascadeClassifier(harcascade)
17     img_gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
18
19     plates = plate_cascade.detectMultiScale(img_gray, 1.1, 5)
20
21     for (x,y,w,h) in plates:
22         area = w * h
23
24         if area > min_area:
25             cv2.rectangle(img, (x,y), (x+w,y+h), (0,255,0), 2)
26             cv2.putText(img, "Found Plate", (x,y-5), cv2.FONT_HERSHEY_COMPLEX_SMALL, 1, (255, 0, 255), 2)
27
28             img_roi = img[y:y+h, x:x+w]
29             cv2.imshow("ROI", img_roi)
30
31     cv2.imshow("Main", img)
32
33     if cv2.waitKey(1) & 0xFF == ord('q'):
34         cv2.imwrite("plate_scanned_img_" + str(count) + ".jpg", img_roi)
35         cv2.rectangle(img, (0,200), (640,200), (0,255,0), cv2.FILLED)
36         cv2.putText(img, "Plat: Disinput", (150, 205), cv2.FONT_HERSHEY_COMPLEX_SMALL, 2, (0, 0, 255), 2)
37         cv2.imshow("hasil", img)
38         cv2.waitKey(500)
39         count += 1
40
41     if cv2.waitKey(1) & 0xFF == ord('s'):
42         cv2.imwrite("hasil_" + str(count) + ".jpg", img)
```

No	Nama File	Plat	Terdeteksi	Hasil
1	scanned_img_0.jpg	BK5403 AGH	Tidak terdeteksi lengkap	BK5403 Agh
2	scanned_img_1.jpg	BK2515 AGP	Tidak terdeteksi lengkap	PK2515 Aef
3	scanned_img_2.jpg	BK4758 AGB	Terdeteksi lengkap	BK4758 AGB
4	scanned_img_3.jpg	B 4150 ML	Tidak terdeteksi lengkap	R 4150ML
5	scanned_img_4.jpg	B 1067 NBF	Tidak terdeteksi lengkap	1067 NBF] 4122
6	scanned_img_5.jpg	B 505 WLG	Tidak terdeteksi lengkap	B 505 WLG 08+27
7	scanned_img_6.jpg	AG 4520 I	Tidak terdeteksi lengkap	Ag 4520
8	scanned_img_7.jpg	AB 3383 XB	Tidak terdeteksi lengkap	AB 9383. XBI
9	scanned_img_8.jpg	AB 5540 WE	Terdeteksi lengkap	AB 5540 WE
10	scanned_img_9.jpg	AB 4577 IX	Tidak terdeteksi lengkap	AB 4577 IX
11	scanned_img_10.jpg	DA 6434 PCN	Terdeteksi lengkap	DA 6437 PCN
12	scanned_img_11.jpg	F 5012 FM	Tidak terdeteksi lengkap	5012, fM
13	scanned_img_12.jpg	B 4792 FCS	Tidak terdeteksi lengkap	B 4792 FCSI
14	scanned_img_13.jpg	B 1964 SSJ	Terdeteksi lengkap	B 1964 SSJ
15	scanned_img_14.jpg	B 1234 WLG	Terdeteksi lengkap	B 1234 WLG

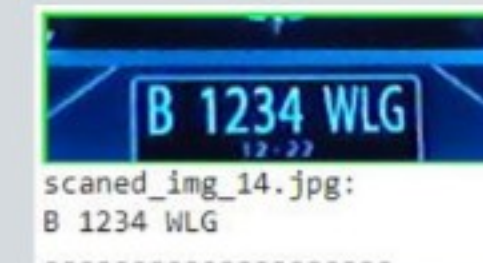


Image: scanned_img_14.jpg
Reference Text: B 1234 WLG
Detected Text: B 1234 WLG
Accuracy: 1.0



Project6

Database for Statistics

Online Bus Ticket Reservation System

Project Overview :

This project designs and implements a database-driven web application for online bus ticket booking. The system enables passengers to reserve tickets conveniently and allows administrators to manage buses, schedules, routes, and transactions efficiently.

Technology & Tools :

- Programming: PHP, HTML, CSS, JavaScript
- Database: MySQL
- Framework & Libraries: Bootstrap, Select2, FontAwesome
- Development Tools: XAMPP (Apache, MySQL, PHPMyAdmin)

Model & Architecture

- Database Design:
- Entities: Passenger, Bus, Driver, Schedule, City, Ticket Reservation, Admin
- Relationships: One-to-many between Passenger → Ticket, Bus → Schedule, etc.
- Normalized tables with primary & foreign keys for integrity
- Feature Engineering (System Modules):
- Admin Module: CRUD for buses, routes, cities, passengers, transactions, and schedule management
- User Module: Online booking, real-time ticket availability, automated price calculation
- Authentication: Admin login/logout with session handling
- Implementation:
- Database schema in MySQL
- Frontend with HTML + Bootstrap
- Backend with PHP (CRUD operations, validation, error handling)



https://github.com/Albani258/Portofolio_Muhammad-Nashiruddin-Al-Bani/tree/main/Project_06_Online_Bus_Ticket_Reservation_System_PHP_MySQL

Project 7

Artificial Neural Networks

Glass Classification Using Learning Vector Quantization (Study of Identification and Classification of Glass Types Based on Refractive Index and Elemental Composition)

Project Overview :

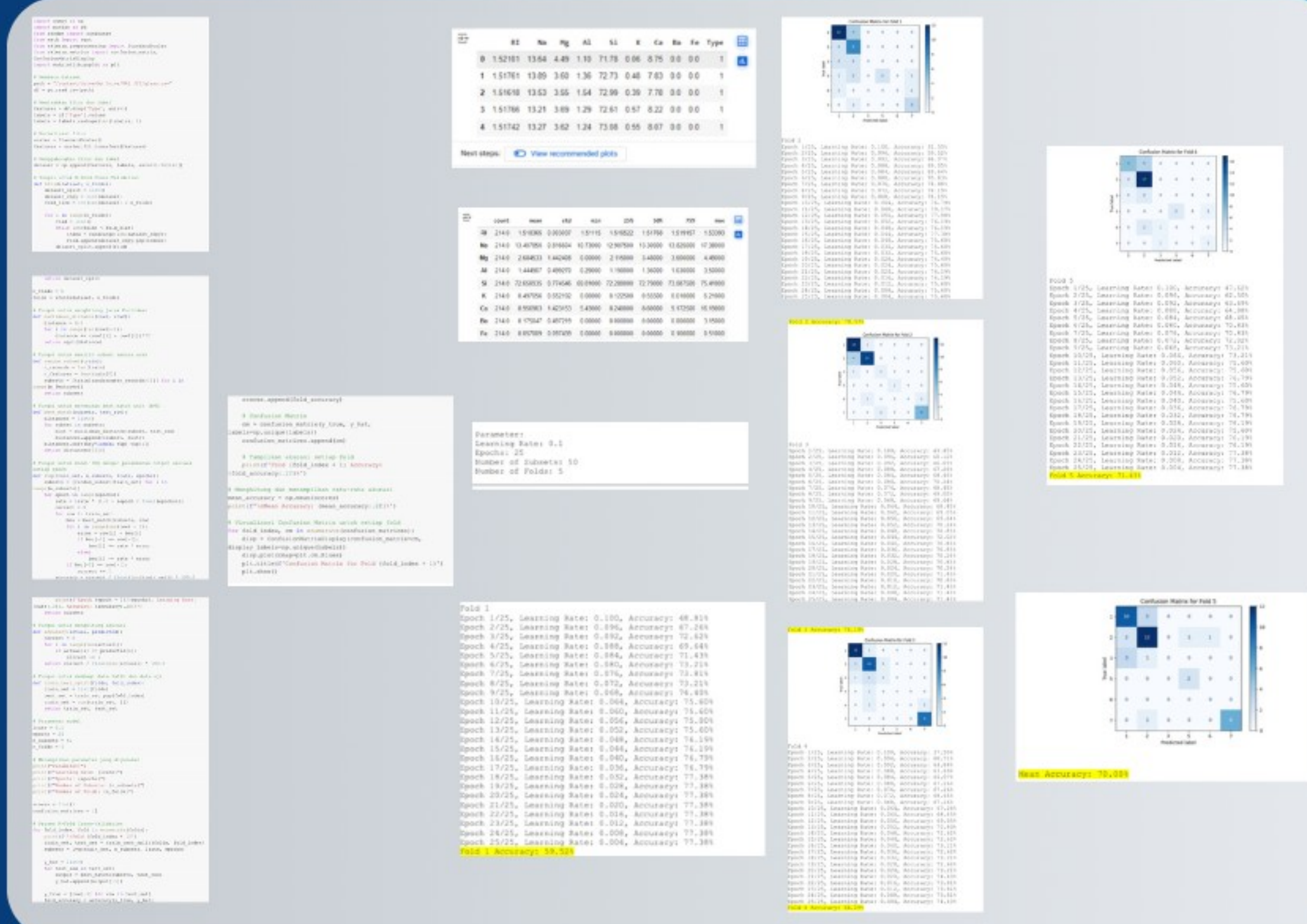
This project focuses on the classification of glass types using Learning Vector Quantization (LVQ), a supervised neural network model. The dataset consists of 214 glass samples with ten numerical features such as refractive index and chemical compositions (Na, Mg, Al, Si, K, Ca, Ba, Fe). The target variable is the type of glass, divided into seven categories (e.g., building windows, vehicle windows, containers, tableware, headlights). The purpose of this study is to support forensic investigation by accurately identifying glass fragments based on their physical and chemical characteristics.

Technology & Tools :

- Programming Language: Python
- Libraries: NumPy, Pandas, Scikit-learn, Matplotlib
- Validation Method: 5-Fold Cross-Validation
- Dataset: UCI Glass Dataset

Model & Architecture

- Machine Learning Model: Learning Vector Quantization (LVQ) neural network for glass type classification.
- Feature Engineering: Data cleaning, normalization, feature-label separation, training with Euclidean distance, learning rate 0.1, 25 epochs, and 50 representative subsets.
- Evaluation: Performance assessed with 5-Fold Cross Validation, confusion matrices, and accuracy metrics, achieving an average accuracy of 70% and peak accuracy of 78.57%.



Project8

Risk Modeling

Optimal Portfolio Analysis in the Energy Sector Stock Using the Markowitz Model

Project Overview :

This study analyzes the optimal portfolio allocation in the Indonesian energy sector using the Markowitz Mean-Variance Model. The analysis is motivated by the global energy crisis triggered by the Russia-Ukraine war, which caused significant price fluctuations in energy-related stocks. Three companies were selected: PT AKR Corporindo Tbk (AKRA), PT Adaro Minerals Indonesia Tbk (ADMR), and PT Medco Energi Internasional Tbk (MEDC), which represent major players in energy distribution, coal, oil, and gas. The objective is to determine the optimal stock weights that maximize return and minimize risk.

Technology & Tools :

- Programming Environment: RStudio
- Packages: tseries, PerformanceAnalytics
- Data Source: Daily closing prices from Yahoo Finance (April 26, 2023 – April 23, 2024)
- Analysis: Expected return, variance, covariance, and standard deviation calculations

Model & Architecture

- Machine Learning Model: Markowitz Mean-Variance Portfolio Optimization
- Feature Engineering: Transformation of daily prices into returns, calculation of expected return, variance, and covariance matrix, formulation of optimization problem using the Lagrange method, and solution in matrix form to obtain portfolio weights.
- Results: The optimal portfolio weights were 56.1% for AKRA, 23.3% for ADMR, and 20.6% for MEDC, with an expected portfolio return of 0.00081 and a portfolio risk (standard deviation) of 0.0428. The results indicate a well-diversified portfolio with stable profit potential and moderate risk.

```
# Load libraries
library(tseries)
library(PerformanceAnalytics)
library(tidyverse)

# Data Source
data_AKRA <- read.csv("https://finance.yahoo.com/ticker/vx/AKRA.JK")
data_ADMR <- read.csv("https://finance.yahoo.com/ticker/vx/ADMR.JK")
data_MEDC <- read.csv("https://finance.yahoo.com/ticker/vx/MEDC.JK")

# Convert dates to POSIXct
data_AKRA$date <- as.POSIXct(data_AKRA$date, tz = "Asia/Jakarta")
data_ADMR$date <- as.POSIXct(data_ADMR$date, tz = "Asia/Jakarta")
data_MEDC$date <- as.POSIXct(data_MEDC$date, tz = "Asia/Jakarta")

# Resample data to daily frequency
data_AKRA <- data.frame(resample(data_AKRA, "D"))
data_ADMR <- data.frame(resample(data_ADMR, "D"))
data_MEDC <- data.frame(resample(data_MEDC, "D"))

# Calculate daily returns
ret_AKRA <- (data_AKRA$close - lag(data_AKRA$close, 1)) / lag(data_AKRA$close, 1)
ret_ADMR <- (data_ADMR$close - lag(data_ADMR$close, 1)) / lag(data_ADMR$close, 1)
ret_MEDC <- (data_MEDC$close - lag(data_MEDC$close, 1)) / lag(data_MEDC$close, 1)
```

```
# Number of securities
ns <- ncol(stocks_xts)

# Getting return (we have taken log return)
ret <- apply(log(stocks_xts), 2, diff)

# Expected return, var, sd
> #daily expected return each stock
> er <- apply(ret, 2, mean)
> print("expected return")
[1] "expected return"
> er
      AKRA_XTS      ADMR_XTS      MEDC_XTS
0.0004252017 0.0011062463 0.0015408965

# Menghitung varians
> variance <- apply(ret, 2, var)
> print("varians")
[1] "varians"
      AKRA_XTS      ADMR_XTS      MEDC_XTS
0.0003404731 0.0011304640 0.0012238145

#daily sd
> std <- apply(ret, 2, sd)
> print("standar deviasi")
[1] "standar deviasi"
      AKRA_XTS      ADMR_XTS      MEDC_XTS
0.01845191 0.03362237 0.03498306

#covariance matrix
> covm <- cov(ret)
> covm
      AKRA_XTS      ADMR_XTS      MEDC_XTS
AKRA_XTS 0.0003404731 0.0001039573 0.0002220796
ADMR_XTS 0.0001039573 0.0011304640 0.0002184089
MEDC_XTS 0.0002220796 0.0002184089 0.0012238145
```

```
> Am <- rbind(2*covm, rep(1, ns))
> Am <- cbind(Am, c(rep(1, ns), 0))
> b <- c(rep(0, ns), 1)
> Am
      AKRA_XTS      ADMR_XTS      MEDC_XTS
AKRA_XTS 0.0006809462 0.0002079146 0.0004441593 1
ADMR_XTS 0.0002079146 0.0022609279 0.0004368177 1
MEDC_XTS 0.0004441593 0.0004368177 0.0024476290 1
      1.0000000000 1.0000000000 1.0000000000 0
> b
[1] 0 0 0 1
```

```
# Optimizing the Markowitz model using the Lagrange method
# Define the objective function
f <- function(w) {
  # Expected return
  er <- apply(ret, 2, mean)
  # Portfolio expected return
  er_p <- sum(w * er)
  # Portfolio variance
  var_p <- sum(w * covm * w)
  # Standard deviation
  sd_p <- sqrt(var_p)
  # Objective function: maximize return, minimize risk
  f <- er_p - lambda * (sd_p - target_sd)
  # Lagrange multiplier
  lambda <- (target_sd - sd_p) / (sd_p^2 - var_p)
  # Optimal weights
  w_opt <- (er - er_p) / (covm - sd_p^2 * covm)
  w_opt <- w_opt / sum(w_opt)
  # Optimal portfolio
  w_opt
}

# Target standard deviation
target_sd <- 0.0428

# Optimal portfolio weights
w_opt <- f(w = rep(1/3, ns))

# Expected return of the optimal portfolio
er_opt <- sum(w_opt * er)

# Standard deviation of the optimal portfolio
sd_opt <- sqrt(sum(w_opt * covm * w_opt))

# Print results
print(paste("Optimal portfolio weights:", w_opt))
print(paste("Expected return of the optimal portfolio:", er_opt))
print(paste("Standard deviation of the optimal portfolio:", sd_opt))
```



Project9

Econometrics Learning Book

A Practical Guide to Creating a Stock Portfolio App with R Shiny

Project Overview :

This project focuses on the development of a web-based financial portfolio application using R Shiny. The application allows users to fetch stock data directly from Yahoo Finance, calculate financial ratios such as ROA, ROE, DER, and PBV, and perform portfolio optimization using the Mean-Variance (MV) Model. The aim is to create an interactive dashboard that combines stock market data visualization, financial ratio analysis, and portfolio optimization in a single platform, making it easier for investors and analysts to monitor and evaluate their portfolios in real time.

Technology & Tools :

- Programming Environment: RStudio
- Framework: R Shiny, Shiny Dashboard
- Packages: quantmod, tidyquant, PerformanceAnalytics, PortfolioAnalytics, ROI, ggplot2, DT, openxlsx
- Data Source: Yahoo Finance API
- Deployment: Shinyapps.io / Shiny Server

Model & Architecture

- Machine Learning Model: Markowitz Mean-Variance Portfolio Optimization (without short sales)
- Feature Engineering: Stock data retrieval from Yahoo Finance, calculation of returns and covariance matrix, financial ratio computation (ROA, ROE, DER, PBV), and statistical description of returns.
- System Architecture:
- User Interface: Interactive dashboard with menus for stock data, financial ratios, and statistical description.
- Server Logic: Fetching and processing stock data, computing returns, running quadratic optimization for MV portfolio, and displaying outputs dynamically.
- Results: The application generates stock return plots, financial ratio tables, descriptive statistics, and optimal portfolio weights visualized through bar charts and tables, enabling users to make data-driven investment decisions.



[https://github.com/Albani258/Portofolio Muhammad-Nashiruddin-Al-Bani/tree/main/Project_09_Stock_Portfolio_App_R_Shiny](https://github.com/Albani258/Portofolio_Muhammad-Nashiruddin-Al-Bani/tree/main/Project_09_Stock_Portfolio_App_R_Shiny)

Project 10

Undergraduate Thesis

ANALYSIS OF BLACK LITTERMAN PORTFOLIO PERFORMANCE MEASUREMENT USING THE STUDENT-T COPULA APPROACH ON IDX30 STOCK

Project Overview :

This research aims to evaluate the performance of investment portfolios by comparing the Original Black-Litterman Model and the Modified Black-Litterman Model with Student-t Copula on IDX30 stocks. The objective is to examine whether the copula-based model provides better efficiency in capturing nonlinear dependencies and tail risk, which are common in volatile financial markets.

Technology & Tools :

- Programming Language: R
- Data Source: Monthly closing prices of IDX30 (May 2016 – September 2024)
- Analysis: CAPM, Johansen cointegration test, Vector Error Correction Model (VECM)
- Evaluation: Sharpe Ratio, Risk & Return metrics

Model & Architecture

- Machine Learning Model: Black-Litterman Portfolio Optimization with Student-t Copula
- Feature Engineering: Data preparation, equilibrium return estimation with CAPM, investor views generated via VECM, dependency modeling using Student-t Copula, portfolio rebalancing with 90-month rolling window, and performance evaluation using return, risk, and Sharpe Ratio.



https://github.com/Albani258/Portofolio_Muhammad-Nashiruddin-Al-Bani/tree/main/Project_10_Undergraduate_Thesis_Black_Litterman_Student_t_Copula_R

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mnashiruddin.albani23@gmail.com

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mnashiruddin.albani23@gmail.com

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mnashiruddin.albani23@gmail.com



Basis Data untuk Statistika

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Analisis Runtun Waktu



Data Mining untuk Statistika



Teknik dan Survei Sampel



Rancangan Percobaan

Contact

- ☎ +62 812-9593-4952
 - in linkedin.com/in/mhmd-al-bani
 - 📍 Bekasi, Jawa Barat, Indonesia
 - ✉ mnashiruddin.albani23@gmail.com
-





THANK YOU

Muhammad Nashiruddin Al Bani

